

Instructor

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Q1. 1.00 mole of a monoatomic ideal gas at 298 K is expanded from 10 m³ to 20 m³ isothermally and reversibly. Calculate ΔS and ΔS_{surr} and ΔH in the process.

Ans:

$$dS_{\text{sys}} = \frac{dq_{\text{rev}}}{T}$$

$$dS_{\text{sys}} = \frac{dU - dw_{\text{rev}}}{T}$$

$$dU = C_v dT$$

For isothermal process $dT = 0$

Thus, $dU = 0$

For a reversible process,

$$dw = -pdV$$

Therefore

$$dS_{\text{sys}} = \frac{dU - dw_{\text{rev}}}{T} = \frac{pdV}{T} = \frac{R}{V} dV$$

$$\Delta S_{\text{sys}} = R \ln \frac{V_2}{V_1} = 5.76 \text{ J mol}^{-1} \text{ K}^{-1}$$

$$dS_{\text{surr}} = \frac{-dq}{T}$$

For a reversible process

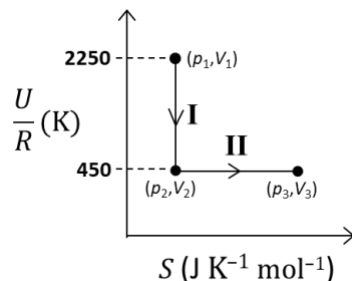
$$dS_{\text{surr}} = \frac{-dq_{\text{rev}}}{T} = -dS_{\text{sys}}$$

$$\Delta S_{\text{surr}} = -\Delta S_{\text{sys}} = -5.76 \text{ J mol}^{-1} \text{ K}^{-1}$$

$$\Delta S_{\text{Total}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} = 0$$

Q2:

- Q.17 One mole of an ideal gas at 900 K, undergoes two reversible processes, **I** followed by **II**, as shown below. If the work done by the gas in the two processes are same, the value of $\ln \frac{V_3}{V_2}$ is ____.



(U : internal energy, S : entropy, p : pressure, V : volume, R : gas constant)

(Given: molar heat capacity at constant volume, $C_{V,m}$ of the gas is $\frac{5}{2}R$)

Ans: 10

Process I is constant entropy process and $dq=0$

$$dw = dU = C_v dT$$

$$\Delta U = (450 - 2250)R = -1800 R = w = C_v (T_2 - T_1) = \frac{5}{2}R(T_2 - 900)$$

$$\begin{aligned} w &= -1800 R \\ (T_2 - 900) &= -720 \\ T_2 &= 180 K \end{aligned}$$

Process II is constant internal energy process and since $dU = C_v dT$, temperature is constant in this process.

$$\begin{aligned} dw &= -pdV = -\frac{RT}{V} dV \\ w &= -RT_2 \ln \frac{V_3}{V_2} \end{aligned}$$

Work done in two processes are same

$$\begin{aligned} -RT_2 \ln \frac{V_3}{V_2} &= -1800 R \\ \ln \frac{V_3}{V_2} &= 10 \end{aligned}$$

Q3. 1.00 mole of a monoatomic ideal gas at 298 K is expanded from 10 m³ to 20 m³ isothermally against a constant external pressure of 10⁵ Pa. Calculate ΔS and ΔS_{surr} , and ΔH in the process.

Ans: $dS_{\text{sys}} = \frac{dq_{\text{rev}}}{T}$

$$dS_{\text{sys}} = \frac{dU - dw_{\text{rev}}}{T}$$

$$dU = C_v dT$$

For isothermal process $dT = 0$

Thus, $dU = 0$

For a reversible process,

$$dw = -pdV$$

Therefore

$$dS_{\text{sys}} = \frac{dU - dw_{\text{rev}}}{T} = \frac{pdV}{T} = \frac{R}{V} dV$$

$$\Delta S_{\text{sys}} = R \ln \frac{V_2}{V_1} = 5.76 \text{ J mol}^{-1} \text{ K}^{-1}$$

$$dS_{\text{surr}} = \frac{-dq}{T}$$

For this process, $dq \neq dq_{\text{rev}}$

$$dS_{\text{surr}} = \frac{dq}{T} = \frac{dU - dw}{T} = \frac{0 - (-P_{\text{ext}} dV)}{T} = \frac{P_{\text{ext}} dV}{T}$$

$$\Delta S_{\text{surr}} = -3355.7 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$dH = C_p dT$$

$$\Delta H = 0$$

Q4. 1.00 mole of a monoatomic ideal gas at 298 K is expanded from 10 m³ to 20 m³ adiabatically and reversibly. Calculate ΔS and ΔS_{surr} , ΔH and ΔT in the process.

Ans: 0, 0, -1371.8 J mol⁻¹, 110 K

For an adiabatic process, $dq = 0$

$$dS_{\text{sys}} = \frac{dq_{\text{rev}}}{T} = 0$$

$$dS_{\text{surr}} = \frac{-dq}{T} = 0$$

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1}$$

For monoatomic gas,

$$C_v = \frac{3}{2}R, C_p = \frac{5}{2}R, \quad \gamma = \frac{C_p}{C_v} = \frac{5}{3}$$

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1}$$

$$T_2 = \left(\frac{1}{2}\right)^{2/3} T_1$$

$$dH = C_p dT$$

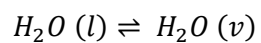
$$\Delta H = C_p(T_2 - T_1)$$

Q5. ΔH of vaporization of water is 539.4 cal/g at the normal boiling point. (a) Many bacteria can survive at 100 °C by forming spores. Hence autoclaves used to sterilize medical and laboratory instruments are pressurized to raise the boiling point of water to 120 °C. What will be the pressure needed to increase the boiling point of water to 120 °C.

Ans: 1.94 atm

$$\ln \frac{K_2}{K_1} = -\frac{\Delta H}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

For



$$K = p$$

$$\ln \frac{p_2}{p_1} = -\frac{\Delta H}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

Q6. On the sea bottom at the Galapagos Rift, water heated to 350 °C gushes out of hydrothermal vents at a depth of 3000 m. Will this water boil or remain liquid at this depth ? The vapor pressure of water is 163 atm at 350 °C.

Ans: The concept behind this has not been discussed in class. So skip this question.

Q7. In a chemical reaction peroxydisulphate ions are used to oxidize iodide ions to triiodide. If rate of disappearance of peroxydisulphate ions is $1.37 \times 10^{-3} \text{ mol.L}^{-1}.\text{s}^{-1}$, calculate the rate with which iodide ions will disappear. Also calculate the rate of formation of sulphate ions.

Solution: We have, $3\text{I}^- + \text{S}_2\text{O}_8^{2-} \rightarrow \text{I}_3^- + 2\text{SO}_4^{2-}$

$$\text{Rate of reaction} = -\frac{1}{3} \frac{d[\text{I}^-]}{dt} = -\frac{d[\text{S}_2\text{O}_8^{2-}]}{dt} = \frac{d[\text{I}_3^-]}{dt} = \frac{1}{2} \frac{d[\text{SO}_4^{2-}]}{dt}$$

$$\text{Given is } -\frac{d[\text{S}_2\text{O}_8^{2-}]}{dt} = 1.37 \times 10^{-3} \text{ mol.L}^{-1}.\text{s}^{-1}$$

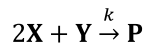
$$\text{Rate of disappearance of } \text{I}^- = -\frac{d[\text{I}^-]}{dt} = -3 \times \frac{d[\text{S}_2\text{O}_8^{2-}]}{dt} = 3 \times 1.37 \times 10^{-3} \text{ mol.L}^{-1}.\text{s}^{-1}$$

$$\text{Rate of disappearance of } \text{I}^- = 4.11 \times 10^{-3} \text{ mol.L}^{-1}.\text{s}^{-1}$$

$$\text{Rate of appearance of } \text{SO}_4^{2-} = \frac{d[\text{SO}_4^{2-}]}{dt} = -2 \times \frac{d[\text{S}_2\text{O}_8^{2-}]}{dt} = 2.74 \times 10^{-3} \text{ mol.L}^{-1}.\text{s}^{-1}$$

Q8.

Q.3 For the following reaction



the rate of reaction is $\frac{d[\text{P}]}{dt} = k[\text{X}]$. Two moles of **X** are mixed with one mole of **Y** to make 1.0 L of solution. At 50 s, 0.5 mole of **Y** is left in the reaction mixture. The correct statement(s) about the reaction is(are)

(Use: $\ln 2 = 0.693$)

(A) The rate constant, k , of the reaction is $13.86 \times 10^{-4} \text{ s}^{-1}$.

(B) Half-life of **X** is 50 s.

(C) At 50 s, $-\frac{d[\text{X}]}{dt} = 13.86 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$.

(D) At 100 s, $-\frac{d[\text{Y}]}{dt} = 3.46 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$.

Ans: B, C and D

Ans:

$$\begin{array}{ccc}
 2\text{X} & + & \text{Y} \longrightarrow \text{P} \\
 2 & & 1 \\
 2-2x & & 1-x
 \end{array}$$

$$1-x = 0.5 \quad ; \quad x = 0.5$$

$$[\text{X}] = \frac{2-2 \times 0.5}{1} = 1$$

Half life of X is 50s.

$$\frac{-d[\text{X}]}{[\text{X}]} = 2k dt \quad ; \quad \ln 2 = 2k t_{1/2}$$

$$k = \frac{\ln 2}{2t_{1/2}} = \frac{0.693}{2 \times 50} = 6.93 \times 10^{-3} \text{ s}^{-1}$$

$$\frac{-d[\text{X}]}{dt} = 2k[\text{X}] = 2 \times 6.93 \times 10^{-3} \times 1 = 13.86 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$$

$$\frac{-d[\text{X}]}{dt} \text{ at } t=50\text{s} = 2 \times 6.93 \times 10^{-3} \times 1 = 13.86 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$$

$$\frac{-d[\text{X}]}{dt} \text{ at } t=100\text{s} = 2 \times 6.93 \times 10^{-3} \times \frac{1}{2} = 6.93 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$$

$$\frac{-d[\text{X}]}{dt} = -\frac{1}{2} \frac{d[\text{X}]}{dt} = \frac{6.93 \times 10^{-3}}{2} = 3.46 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$$

Q9. The rate constant for a zero order reaction $A \rightarrow \text{Products}$ is $2.1 \times 10^{-2} \text{ mol L}^{-1} \text{ s}^{-1}$. If the initial concentration of A is 21 moles per litre, calculate its concentration after 15 minutes.

Solution: The integrated rate expression for a zero order reaction is

$$kt = [A_0] - [A]$$

$$[A] = [A_0] - kt = 21 - (2.1 \times 10^{-2} \times 15 \times 60) = 21 - 18.9 = 2.1 \text{ mol L}^{-1}$$

Q10. The initial concentration of N_2O_5 in the following reaction is $2.37 \times 10^{-3} \text{ mol L}^{-1}$ at 298 K.



Calculate the rate constant for the reaction if the concentration of N_2O_5 is reduced to $0.38 \times 10^{-3} \text{ mol L}^{-1}$ after 50 mins.

Solution: For a first order reaction,

$$\log \frac{[N_2O_5]_{t^o}}{[N_2O_5]_t} = \frac{k(t - t^o)}{2.303}$$

$$k = \log \frac{[N_2O_5]_{t^o}}{[N_2O_5]_t} \times \frac{2.303}{(t - t^o)} = \log \frac{2.37 \times 10^{-3}}{0.38 \times 10^{-3}} \times \frac{2.303}{(50 - 0)} = 0.037 \text{ min}^{-1}$$

Q11. The rate constant for a first order reaction is $1.75 \times 10^{-2} \text{ s}^{-1}$ at 298 K. Calculate the time required to reduce the initial amount of reactant from 10 g to 4 g.

Solution: Initial concentration = 10 g, Final concentration = 4 g, $k = 1.75 \times 10^{-2} \text{ s}^{-1}$

$$t = \log \frac{[A]_o}{[A]} \times \frac{2.303}{k} = \log \frac{10}{4} \times \frac{2.303}{1.75 \times 10^{-2}} = 52.37 \text{ s}$$

Q12. For a chemical reaction: $A + B \rightarrow \text{Products}$, an experiment was performed with three different conditions. The observations for the same are tabulated as:

Exp.	Initial [A] (mol/L)	Initial [B] (mol/L)	Init. Rate of Formation of products (M s^{-1})
1	0.05	0.05	7.8×10^{-4}
2	0.10	0.05	1.56×10^{-3}
3	0.10	0.025	7.8×10^{-4}

Determine the rate law for this reaction and the rate constant. What will be the initial rate of reaction when $[A]_0 = 0.132 \text{ M}$ and $[B]_0 = 0.012 \text{ M}$

Solution: On examining experiments 2 and 3, it is found that A remains constant while B is doubled in concentration which results in doubling the rate of the reaction. Therefore, the reaction is first order in B.

Also, if we look at experiments 1 and 2, we found that B remains constant while the concentration of A doubles. As a result of the doubled concentration, the rate also doubles. Therefore, the reaction is first order in A also.

Hence, the rate law for this reaction is:

$$\text{rate} = k [A] [B]$$

Calculation of the rate constant:

$$\text{rate} = k [A] [B]$$

$$7.8 \times 10^{-4} \text{ M s}^{-1} = k (0.05 \text{ M}) (0.05 \text{ M})$$

$$k = 0.312 \text{ M}^{-1} \text{ s}^{-1}$$

When $[A]_0 = 0.132 \text{ M}$ and $[B]_0 = 0.012 \text{ M}$

$$\text{rate} = (0.312 \text{ M}^{-1} \text{ s}^{-1}) (0.132 \text{ M}) (0.012 \text{ M})$$

$$\text{rate} = 4.94 \times 10^{-4} \text{ M s}^{-1}$$

Q13. In a flask containing inert gas, PH_3 gas is poured at 500°C . The decomposition of PH_3 gas took place into $\text{P}_4(g)$ and $\text{H}_2(g)$ and the total pressure is observed as a function of time.

Time (sec)	0	60	120	∞
P (mm)	246.4	263.9	267.2	268.1

What is the order and rate constant of the reaction?

Solution: Total increase in Pressure due to complete decomposition of $\text{PH}_3 \propto$ Initial conc. of PH_3

$$\text{i.e. } a \propto P_\infty - P_o \text{ or } a \propto (268.1 - 246.4)\text{mm} \quad \text{i.e. } a \propto 21.7 \text{ mm}$$

increase in pressure at any time $\propto$ amount decomposed at that time, i.e. $P_t - P_o \propto x$

$$\therefore (a - x) \propto (P_\infty - P_o) - (P_t - P_o) \text{ or } (a - x) \propto (P_\infty - P_t)$$

If reaction is of first order,

$$k = \frac{2.303}{t} \log \frac{a}{(a - x)} = \frac{2.303}{t} \log \frac{(P_\infty - P_o)}{(P_\infty - P_t)}$$

at $t = 60$ s,

$$k = \frac{2.303}{60} \log \frac{(268.1 - 246.4)}{(268.1 - 263.9)} = 0.0274 \text{ s}^{-1}$$

at $t = 120$ s,

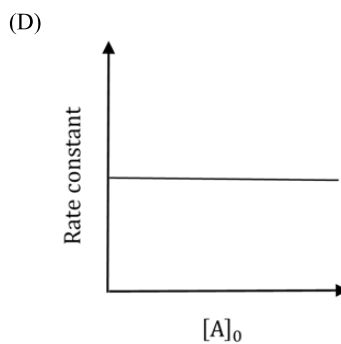
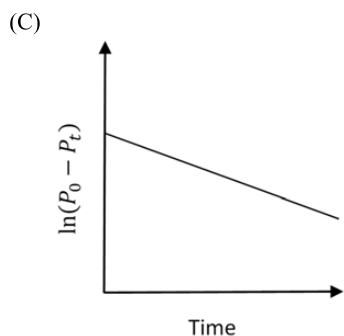
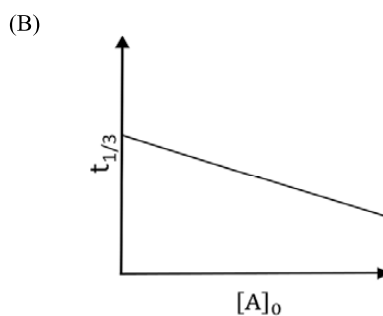
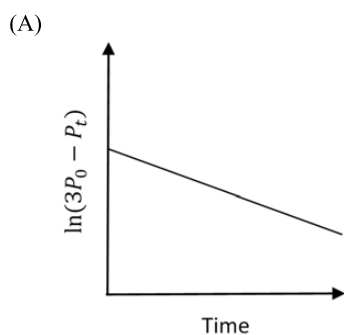
$$k = \frac{2.303}{120} \log \frac{(268.1 - 246.4)}{(268.1 - 267.2)} = 0.0265 \text{ s}^{-1}$$

As k comes out to be nearly constant, it is a reaction of first order.

$$\text{Mean value of } k = \frac{0.0274 + 0.0265}{2} = 0.027 \text{ s}^{-1}$$

Q14.

- Q.5 For a first order reaction $A(g) \rightarrow 2B(g) + C(g)$ at constant volume and 300 K, the total pressure at the beginning ($t = 0$) and at time t are P_0 and P_t , respectively. Initially, only A is present with concentration $[A]_0$, and $t_{1/3}$ is the time required for the partial pressure of A to reach $1/3^{\text{rd}}$ of its initial value. The correct option(s) is (are)
(Assume that all these gases behave as ideal gases)



Ans: A and D



$$x \propto \frac{P_t - P_0}{2} \quad a-x \propto \frac{3P_0 - P_t}{2}$$

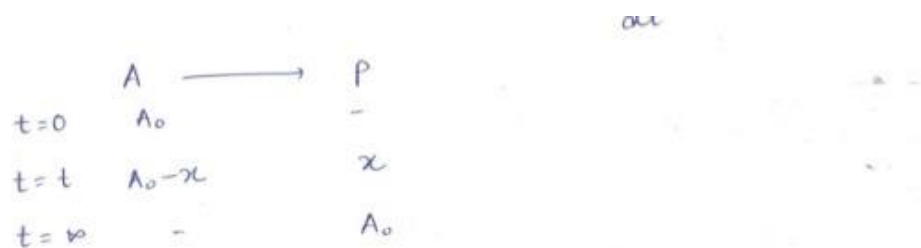
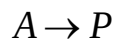
$$\ln \frac{a}{a-x} = \ln \frac{2P_0}{3P_0 - P_t} = kt$$

option A is right, option C is wrong.

$\ln \frac{[A]_0}{[A]} = \ln 3 = k t_{1/3}$; $t_{1/3}$ is independent of $[A]_0$, option B is wrong.

Rate constant is always independent of $[A]_0$.

Q15. Derive the relationship between Y_0 , Y_∞ , Y_t , A_0 , t and rate constant k for a reaction given below, where reaction is of order $\frac{1}{2}$ w.r.t. A.



$$Y_0 = k_1 A_0$$

$$Y_t = k_1 (A_0 - x) + k_2 x$$

$$Y_\infty = k_2 A_0$$

$$\begin{aligned} Y_t - Y_\infty &= k_1 A_0 - k_1 x + k_2 x - k_2 A_0 \\ &= (k_1 - k_2) (A_0 - x) \end{aligned}$$

$$\begin{aligned} Y_0 - Y_\infty &= k_1 A_0 - k_2 A_0 \\ &= (k_1 - k_2) A_0 \end{aligned}$$

$$\therefore \frac{Y_t - Y_\infty}{Y_0 - Y_\infty} = \frac{A_0 - x}{A_0} = \frac{[A_t]}{[A_0]} \quad \text{--- (1)}$$

given $-\frac{d[A]}{dt} = k[A]^{1/2}$

integrating both the sides & putting limits

$$\int_{A_0}^{A_t} \frac{d[A]}{[A]^{1/2}} = -k \int_0^t dt$$

$$2[A]^{1/2} \Big|_{A_0}^{A_t} = -k t \Big|_{t=0}^t$$

$$2 \left[[A_t]^{1/2} - [A_0]^{1/2} \right] = -kt$$

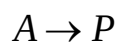
$$2 \left[[A_0]^{1/2} - [A]_t^{1/2} \right] = kt$$

$$2[A_0]^{1/2} \left[1 - \left(\frac{[A]_t}{[A_0]} \right)^{1/2} \right] = kt$$

from eqⁿ (i)

$$\boxed{2[A_0]^{1/2} \left[1 - \left(\frac{Y_t - Y_\infty}{Y_0 - Y_\infty} \right)^{1/2} \right] = kt}$$

Q16. Derive the relationship between Y_0 , Y_∞ , Y_t , A_0 , t and rate constant k for a reaction given below, where reaction is of order 2 w.r.t. A.



Ans.
$$\frac{Y_\infty - Y_t}{Y_\infty - Y_0} = \frac{[A]}{[A]_0}$$

For a second order reaction

$$-\frac{d[A]}{dt} = k[A]^2$$

$$-\int_{[A]_0}^{[A]} \frac{d[A]}{[A]^2} = k \int_{t=0}^{t=t} dt$$

On integration

$$\frac{1}{[A]} - \frac{1}{[A]_0} = kt$$

$$\frac{[A]_0}{[A]} - 1 = kt[A]_0 \quad \left(\begin{array}{l} \text{Multiplying} \\ \text{both side} \\ \text{by } [A]_0 \end{array} \right)$$

$$\frac{Y_\infty - Y_t}{Y_\infty - Y_0} = kt[A]_0 + 1$$

$$\frac{Y_0 - Y_\infty}{Y_t - Y_\infty} = kt[A]_0 + 1$$

Q17. The reaction *cis*-2-butene to *trans*-2-butene is first order in both directions. At 25°C, the equilibrium constant is 0.406 and the forward rate constant is $4.21 \times 10^{-4} \text{ s}^{-1}$. Starting with a sample of pure *cis* isomer with $[cis]_0 = 0.115 \text{ mol.dm}^{-3}$, how long would it take for half the equilibrium amount of the *trans* isomer to form?

Ans:

$$\ln \frac{[cis]_0 - [cis]_{eq}}{[cis] - [cis]_{eq}} = (k_f + k_b)t$$

$$[cis]_{eq} + [trans]_{eq} = [cis] + [trans] = [cis]_0$$

From question

$$[trans] = \frac{[trans]_{eq}}{2}$$

$$[cis] = [cis]_0 - \frac{[trans]_{eq}}{2} = [cis]_0 - \frac{[cis]_0 - [cis]_{eq}}{2} = \frac{[cis]_0 + [cis]_{eq}}{2}$$

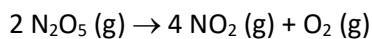
$$[cis] - [cis]_{eq} = \frac{[cis]_0 + [cis]_{eq}}{2} - [cis]_{eq} = \frac{[cis]_0 - [cis]_{eq}}{2}$$

$$\ln 2 = (k_f + k_b)t$$

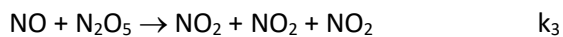
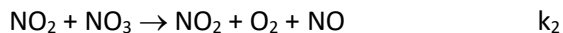
$$k_b = \frac{K}{k_f}$$

$$t = 490 \text{ s}$$

Q18. Devise the rate law for the composition of N_2O_5 :



On the basis of the following mechanism:



Ans:

$$-\frac{d[N_2O_5]}{dt} = k_1[N_2O_5] - k_{-1}[NO_2][NO_3] + k_3[NO][N_2O_5] \quad \text{---(i)}$$

Applying SSA to $[NO_3]$

$$\frac{d[NO_3]}{dt} = k_1[N_2O_5] - k_{-1}[NO_2][NO_3] - k_2[NO_2][NO_3] \approx 0$$

$$[NO_2][NO_3] = \frac{k_1[N_2O_5]}{(k_{-1} + k_2)} \quad \text{---(ii)}$$

Applying SSA to $[NO]$

$$\frac{d[NO]}{dt} = k_2[NO_2][NO_3] - k_3[NO][N_2O_5]$$

$$k_3[NO][N_2O_5] = k_2[NO_2][NO_3] \quad \text{---(iii)}$$

From (i) and (iii)

$$-\frac{d[N_2O_5]}{dt} = k_1[N_2O_5] - k_{-1}[NO_2][NO_3] + k_2[NO_2][NO_3]$$

$$= k_1[N_2O_5] + (k_2 - k_{-1})[NO_2][NO_3]$$

$$= k_1[N_2O_5] + \frac{(k_2 - k_{-1}) k_1}{k_{-1} + k_2} [N_2O_5]$$

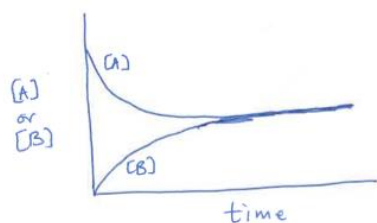
$$= \frac{(k_{-1} + k_2 + k_2 - k_{-1}) k_1}{k_{-1} + k_2} [N_2O_5]$$

$$= \frac{2k_1k_2}{k_{-1} + k_2} [N_2O_5]$$

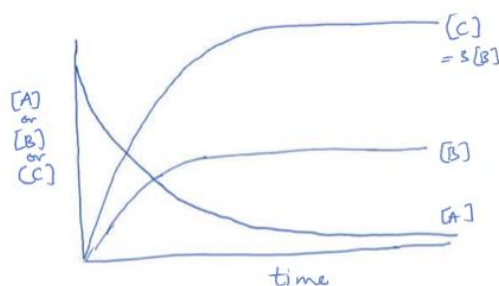
vQ19. Plot the fraction present of each species versus the time elapsed for the following cases:

- (i) A in equilibrium with B (first order in both directions, $[A]_0 \neq 0$ and $[B]_0 = 0$), with $k_1/k_{-1} = 1$.
- (ii) A simultaneously decomposing into B and C with first-order rate constants k_1 and k_2 , respectively, and $k_2/k_1 = 3$ ($[A]_0 \neq 0$, $[B]_0 = 0$, and $[C]_0 = 0$).
- (iii) A in equilibrium with B and C simultaneously with first-order rate constants $\{k_1, k_{-1}\}$ and $\{k_2, k_{-2}\}$, respectively, and $k_1 = 1 \times 10^{-2}$, $k_{-1} = 1 \times 10^{-4}$, $k_2 = 1 \times 10^{-3}$, and $k_{-2} = 1 \times 10^{-7}$ (with proper units).
- (iv) A decaying to B and B decaying to C with first-order rate constants of k_1 and k_2 , respectively, and $k_1/k_2 = 20$ ($[A]_0 \neq 0$, $[B]_0 = 0$, and $[C]_0 = 0$).
- (v) A decaying to B and B decaying to C with first-order rate constants of k_1 and k_2 , respectively, and $k_2/k_1 = 20$ ($[A]_0 \neq 0$, $[B]_0 = 0$, and $[C]_0 = 0$).

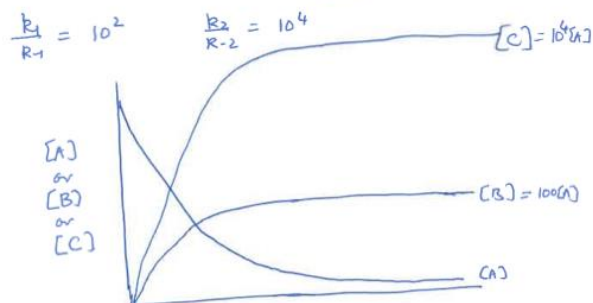
(i)



(ii)



(iii)



(iv)

